

JAVASCRIPT OBJECTS, NESTED OBJECTS & JSON ASSIGNMENT

1. Create a Simple Object

Create a JavaScript object named `student` with the following properties:

name
age
course
city

Use the following data:

Name: Rahul
Age: 25
Course: Playwright Automation
City: Delhi

Print all the property values using console.log().

2. Access Object Properties

Given:

```
let employee = {  
  name: "John",  
  age: 30,  
  department: "QA",  
  salary: 75000  
};
```

Print the following individually:

Employee Name:
Employee Age:
Employee Department:
Employee Salary:

Access the properties using dot notation.

3. Access Properties Using Bracket Notation

Using the same `employee` object, access all properties using bracket notation.

For example:

```
employee["name"]
```

4. Add New Properties

Given:

```
let employee = {  
  name: "John",  
  age: 30,  
  department: "QA"  
};
```

Add the following properties:

```
company: "ABC Technologies"  
experience: 6  
location: "Bangalore"
```

Print the updated object.

5. Modify Object Properties

Given:

```
let employee = {  
  name: "John",  
  age: 30,  
  salary: 60000
```

```
};
```

Perform the following:

1. Change age to 31.
2. Change salary to 75000.
3. Add designation: "Automation Tester".

Print the updated object.

6. Delete Object Properties

Given:

```
let employee = {  
  name: "John",  
  age: 30,  
  department: "QA",  
  salary: 75000  
};
```

Remove the salary property using the delete operator.

Print the resulting object.

7. Create a Nested Object

Create an object named `employee` with the following structure:

```
let employee = {  
  name: "John",  
  age: 30,  
  address: {  
    city: "Bangalore",  
    state: "Karnataka",  
    country: "India"  
  }  
};
```

Print:

Employee Name

City

State

Country

8. Access Nested Object Properties

Given:

```
let student = {  
  name: "Rahul",  
  course: {  
    name: "Playwright Automation",  
    duration: "3 Months",  
    mode: "Online"  
  }  
};
```

Print:

Student Name:

Course Name:

Course Duration:

Course Mode:

9. Modify Nested Object Properties

Given:

```
let employee = {  
  name: "John",  
  address: {  
    city: "Delhi",  
    state: "Delhi"  
  }  
}
```

```
};
```

Change:

city → Mumbai

state → Maharashtra

Print the updated object.

10. Add Properties to a Nested Object

Given:

```
let employee = {  
  name: "John",  
  address: {  
    city: "Delhi"  
  }  
};
```

Add:

```
state: "Delhi"  
country: "India"  
pincode: 110001
```

Print the complete object.

11. Multi-Level Nested Object

Create an object with the following structure:

```
let employee = {  
  name: "John",  
  company: {  
    name: "ABC Technologies",  
    location: {  
      city: "Bangalore",
```

```
        state: "Karnataka",
        country: "India"
    }
}
};
```

Print:

Employee Name
Company Name
City
State
Country

12. Nested Object with Multiple Objects

Create the following object:

```
let employee = {
  name: "John",
  personalDetails: {
    age: 30,
    gender: "Male"
  },
  professionalDetails: {
    designation: "Automation Tester",
    experience: 6
  }
};
```

Print all the values from the object.

13. Identify JSON vs JavaScript Object

Identify whether each of the following is a JavaScript Object or JSON.

A.

```
let employee = {  
  name: "John",  
  age: 30  
};
```

B.

```
{  
  "name": "John",  
  "age": 30  
}
```

Explain the difference between the two.

14. Convert JSON String to JavaScript Object

Given the following JSON string:

```
let jsonData = '{"name":"John","age":30,"department":"QA"}';
```

Convert it into a JavaScript object using:

```
JSON.parse()
```

Then print:

Name

Age

Department

15. Convert JavaScript Object to JSON

Given:

```
let employee = {  
  name: "John",  
  age: 30,  
  department: "QA"
```

```
};
```

Convert the JavaScript object into a JSON string using:

```
JSON.stringify()
```

Print the result.